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Dementia Therapy: The Role of Gamified AI and Digital Art in Supporting Cognitive and Emotional Well-being

Pezhman Raeisian Parvari[a,*], Damian Schofield[b]

[a]*School of Medicine, Indiana University, Indianapolis, USA*
[b]*Department of Computer Science, State University of New York, Oswego, New York,, USA*

Abstract

Millions of people worldwide are affected by dementia, a cognitive health condition that has a major negative influence on memory, judgment, and overall quality of life. Due to accessibility and cost issues, conventional treatments require creative solutions. For those with dementia, art therapy provides a creative means of increasing their mental and emotional health. These treatments are further improved by emerging technologies like augmented reality (AR), virtual reality (VR), and artificial intelligence (AI), which provide engaging, personalized, and appealing experiences. This study investigates how dementia-related cognitive impairments may be addressed by using AI into art therapy.

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Keywords: Dementia; Art Therapy; Digital Art Therapy; Artificial Intelligence; Gamified Activities; Game; Assistive Technology

* Corresponding author.
E-mail address: pezhmanraeisian@yahoo.com

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1. Introduction

According to the World Health Organization (WHO), there are about 50 million dementia sufferers globally; by 2030, that figure is expected to increase to 82 million, and by 2050, it will reach 152 million [1]. With rapid advancements in technology, this growing population is becoming an important focus for targeted innovations. However, given their unique needs, it is crucial to develop technologies tailored to address each individual's requirements.

In [2], the authors emphasize the importance of interventions aimed at preventing the process of Alzheimer's disease, a prevalent kind of dementia. However, obstacles including the expense of therapy, limited access in some areas, and the strain of commuting to therapies frequently make it difficult for clients to receive treatments [3]. These barriers highlight the importance of designing accessible and effective Art Therapy interventions for this population. Digital art therapy has demonstrated potential in addressing the needs of people with dementia [3]. Additionally, research has shown the possible benefits of games for dementia patients [4]. Building on previous discoveries, this paper explores how gamified activities, supported by emerging technologies like artificial intelligence (AI), can offer personalized and engaging experiences for people with dementia.

2. Dementia

The CDC states that dementia is a general term that refers to a deterioration in cognitive functions, including memory, thinking, and decision-making, that substantially disrupts day-to-day functioning rather than a specific disease and the most common type of dementia is Alzheimer's disease [5]. Memory, communication, executive functioning, and decision-making abilities are among the cognitive abilities that are impaired in people with dementia [6]. People's quality of life is significantly impacted by the degree and stages of dementia, which frequently diminish their independence and increase feelings of loneliness [7,8]. To address the complex demands of people with dementia, a comprehensive model has been created [3] (see Figure 1). The individual is at the center of this model, which looks at every facet of their everyday lives, from simple to complex actions. By examining the interactions between obstacles, feelings, and daily activities experienced by individuals with dementia, this method aims to offer a framework for comprehending and meeting their particular requirements.

3. Art Therapy and People with Dementia

According to the American Art Therapy Association, art therapy is a practice which employs techniques that go beyond spoken language to integrate the mind, body, and spirit. By overcoming the limitations of language, these methods may motivate people while providing visual and symbolic representations to express individual and collective experiences [9]. Furthermore, creative projection, the triangle interaction, embodiment, nonverbal communication, playful exploration, and the active roles of the therapist and witness are some of the key concepts that authors in [10] sees as supporting art therapies. These guidelines emphasize the therapeutic value and intricacy of creating art [10]. Through creative processes and results, art therapy encourages self-expression both verbally and nonverbally.

Digital art therapy (RT) has shown significant benefits in improving cognitive function and enhancing the quality of life for clients [3]. Several case studies and examples demonstrate the positive impact of digital art therapy in dementia care. Studies indicate that RT, particularly digital life stories and networked RT, helps to reduce symptoms of depression and agitation while improving memory recall and emotional expression. For instance, digital life storybooks have been useful for individuals with mild to moderate dementia in evoking positive emotions, improving depressive symptoms, and recalling cherished memories [11]. These therapies can be more effective than traditional methods, especially when it comes to alleviating cognitive and emotional symptoms [12,13]. The use of technology in art therapy has been found to benefit both individuals and their caregivers, offering a cost-effective and efficient method for promoting emotional and cognitive well-being [14, 15, 3]. Another notable case is the use of social robots,

like PARO, a baby harp seal robot, which has been developed as a therapeutic tool for PWD. PARO, powered by artificial intelligence, responds to its environment and users, providing interaction that mimics pet therapy [16].

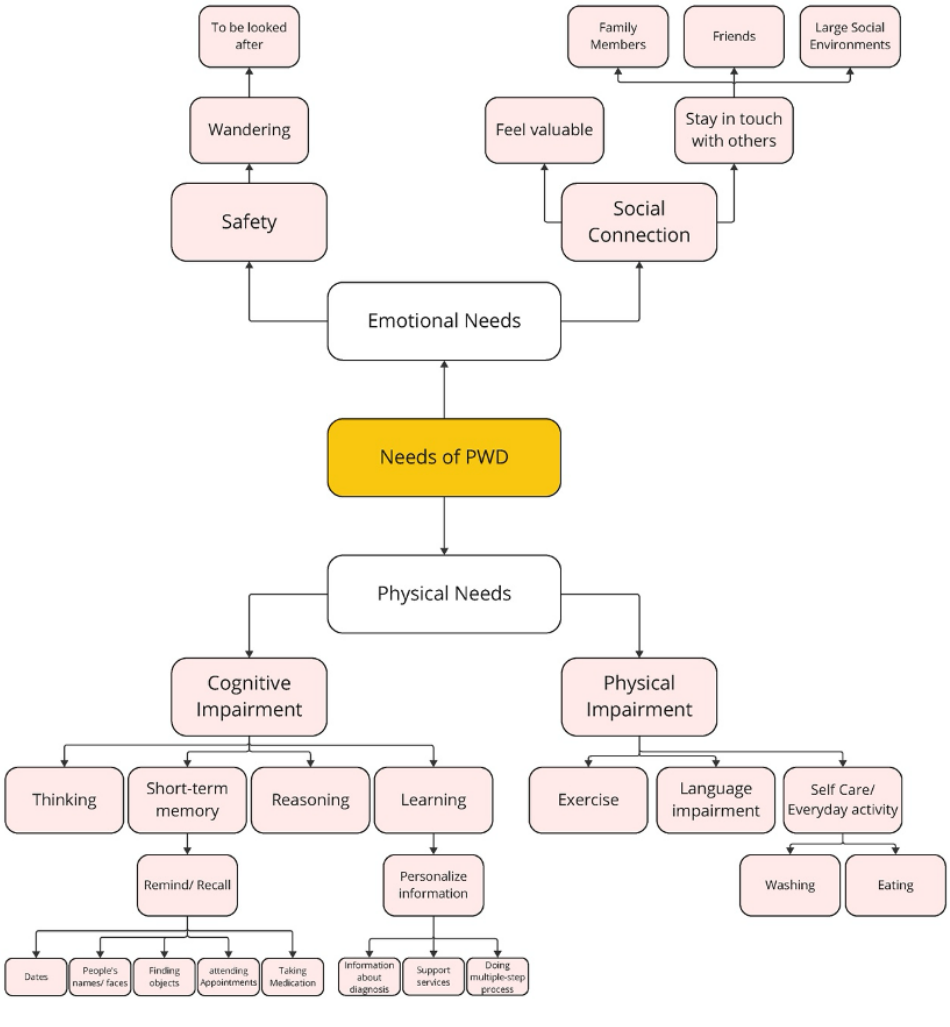


Fig. 1. Model of Needs of People with Dementia [3]

Virtual reality (VR), augmented reality (AR), and digital platforms like tablets, have transformed art therapy by enabling immersive and interactive experiences that engage PWD in novel ways. VR, for example, has been shown to improve attention and memory functions in people with dementia [17]. VR simulations allow users to interact with virtual environments through tools like VR gloves and head-mounted displays, creating a sense of immersion that traditional therapies cannot provide [18]. Additionally, VR technologies have been effectively utilized for cognitive training, diagnosis, and maintenance of dementia-related conditions, further supporting the potential of immersive experiences for individuals with dementia [19]. In one example, the use of VR-based cognitive training has been shown to enhance cognitive and motor skills in dementia patients, with virtual environments providing both visual and auditory stimuli that help simulate real-world experiences [20]. These interventions not only improve cognitive abilities but also help individuals manage symptoms such as anxiety and agitation, thus promoting overall well-being [21]. By using these advanced technologies, therapists can create environments that promote memory retention and

emotional well-being in a way that is both engaging and therapeutic [22]. Moreover, VR helps reduce agitation and enhances the independence of individuals with dementia, as it offers cognitive stimulation and calming experiences [23]. The success of VR in improving dementia care underscores its potential for broader application in therapeutic settings [23].

AI applications in art therapy, on the other hand, show promise in supporting vulnerable populations, including individuals with dementia. AI can create customized, adaptive therapeutic interventions that cater to the specific needs of patients with cognitive impairments. This could involve AI-generated art prompts, AI-assisted drawing tools, storytelling prompts, or games that stimulate creativity while accommodating cognitive challenges. Furthermore, AI can assist in monitoring patients' cognitive states over time, tracking changes in their ability to engage in art-related activities. Such capabilities are vital in ensuring that the therapy is suitable and effective for patients with dementia, especially given the complexity and variability of their conditions [24]. These AI-driven tools can help users express themselves through art, offering custom suggestions based on individual preferences, mood, or emotional state. The integration of AI allows for dynamic, individualized art interventions, which is particularly helpful in contexts like art therapy for mental health or cognitive impairments. By continuously monitoring user interaction with art creation tools, AI can identify patterns and trends, offering caregivers quantifiable data about engagement levels and emotional expression [25]. This data helps caregivers understand how their patients are responding to treatment, enabling them to adjust their approach or introduce new techniques to improve therapeutic outcomes. Furthermore, AI-based platforms can synthesize user inputs and provide caregivers with real-time feedback, making caregiving more informed and tailored [26]. Ethical considerations, such as ensuring patient privacy and protecting vulnerable individuals from potential harm, are essential when deploying AI tools in this context [27,28,3].

4. Discussion

The integration of digital technologies in dementia care presents a multifaceted opportunity to improve patient outcomes, especially through gamified challenges, digital art activities, and AI-driven tools. In particular, these technologies can address critical gaps in current dementia care, which often lacks personalized and engaging interventions that focus on both cognitive and emotional well-being. The use of gamified learning and digital art activities can offer personalized, enjoyable, and stimulating experiences for patients, promoting not only cognitive function but also emotional expression and social interaction [3]. As noted by [29], these technologies align with the broader goal of supporting dementia patients in maintaining their functioning, improving leisure and activity engagement, and enhancing caregiving experiences. Further, the potential for AI tools and augmented reality (AR), in reminiscence therapy provides innovative ways to support both patients and caregivers in more personalized, interactive ways [30,31]. Combining gamified challenges with digital art activities provides a hybrid approach that can address multiple needs within dementia care. Digital art activities, for instance, offer a medium for creative expression, fostering emotional well-being and social interaction [32,3]. These activities can reduce the stigma associated with cognitive decline, as they allow patients to engage in non-verbal forms of communication.

According to previous studies, the value of digital art therapy activities to addressing the demands of dementia patients are highlighted in Table 1 below [3]:

Table 1: The connection between digital art therapy and needs for people with dementia

Needs	How digital art therapy helps
Safety	No wandering as there is no need to commute to a place where art therapy takes place.
Social connection	Creating a relationship with an art therapist; doing group therapy fosters social interaction and enhances social skills.
Short-term memory	Improving cognitive and sensory-motor function; increasing independence.
Thinking	Promoting insights; reducing and resolving conflicts.
Learning and reasoning	Fostering self-awareness and cultivating emotional resilience.
Daily activity	Increasing independence through engaging activities.
Language impairment	Creating opportunities for non-verbal communication.

Exercise	Enabling art creation with small gestures, increasing physical independence.

Table 1 indicates that digital art activities improve social connection and give dementia patients a creative medium to express their feelings. Digital art therapy enhances cognitive, emotional, and physical wellness by enabling nonverbal communication and granting autonomy in engagement [3].

Similarly, video game theories offer significant potential for addressing dementia-related challenges. Gamified challenges can stimulate cognitive engagement by creating enjoyable and challenging tasks that support the retention and application of skills [33]. This combination of gamification and art offers an approach that not only improves cognitive and functional abilities but also addresses emotional and social health, which are often secondary considerations in traditional digital interventions for dementia [32]. Table 2 illustrates how theories like Self-Determination Theory (SDT), Flow Theory, Mood Management Theory, and Eudaimonia Theory address key needs for people with dementia [4].

Table 2: How Game Theories Address the Needs of People with Dementia (Content from [4], table is created by authors)

Needs	Self-Determination Theory (SDT)	Flow Theory	Mood Management Theory	Eudaimonia Theory
Safety	Fosters relatedness and enjoyment	Provides balanced experiences	-	Promotes personal growth and enjoyment
Social connection	Encourages relatedness and meaningful relationships	-	-	Enhances relationships and feelings of being valued
Short-term memory	Promotes competence and autonomy	Improves recall through balanced challenges	Reduces stress and improves mood	-
Thinking	-	Reduces stress through balanced challenges	Reduces stress and encourages positive thinking	-
Learning & reasoning	Promotes competence and autonomy	Supports learning through balanced challenges	Improves well-being and reasoning	-
Language impairment	-	Enhances communication through meaningful challenges	-	Improves communication through meaningful entertainment
Exercise	Promotes physical exercise through relatedness and competence	Encourages physical activity through challenges	Promotes exercise through stress relief	-

Interdisciplinary approaches in dementia therapy design are essential to creating effective, patient-centered interventions. By bringing together expertise from fields such as game design, art therapy, and AI development, innovative solutions can be developed that bridge the existing gaps in dementia care. For example, co-creation practices, where both caregivers and dementia patients contribute to the design of mHealth tools, allow for solutions that are more personalized and responsive to evolving needs [34]. Such interdisciplinary collaboration can promote the development of tools that are both technologically sophisticated and deeply attuned to the emotional and practical needs of dementia patients and their families [35,36].

The potential for co-creation using AI tools is significant in dementia care, particularly in the context of reminiscence therapy. AI-driven tools can allow dementia patients and their caregivers to actively participate in therapeutic activities, fostering a deeper connection and a sense of agency. When combined with the principles from Table 2, AI tools can incorporate personalized, well-balanced challenges inspired by Flow Theory or leverage SDT to support autonomy and competence during therapy sessions. Co-creation with caregivers ensures that digital tools are not only useful but also relevant and meaningful to the daily lives of patients. By involving caregivers in the design process, co-creation addresses the challenge of slow adoption of digital tools due to technological literacy issues [34]. Moreover, it ensures that interventions evolve with the progression of the disease, offering personalized and adaptive support over time [35]. This process aligns with the aim of creating more integrated, user-centric care models that enhance both the caregiving experience and the quality of life for dementia patients [34].

Combining the insights from Tables 1 and 2, we propose Table 3 as a conceptual framework for hybrid interventions that integrate digital art therapy and gamified challenges.

Table 3. How Gamified AI Art Therapy Can Address Dementia Needs

Needs	Digital Art Therapy	Gamified Activities	Hybrid Approach (the use of gamified AI activity in digital art therapy)
Safety	No physical wandering required	Provides balanced and secure challenges	Offers safe, structured activities in familiar settings
Social connection	Enhances relationships through group therapy	Fosters relatedness and meaningful relationships (SDT)	Encourages creative collaboration and connection
Short-term memory	Improves sensory-motor function and independence	Promotes recall through competence and autonomy (SDT, Flow)	Combines memory stimulation with creative expression
Thinking	Promotes insights and conflict resolution	Reduces stress through balanced challenges (Flow, Mood)	Stimulates thinking through engaging, creative tasks
Learning & reasoning	Cultivates self-awareness and emotional resilience	Enhances learning through balanced tasks (SDT, Flow)	Combines learning with emotional expression
Language impairment	Facilitates non-verbal communication	Improves communication skills through meaningful challenges	Encourages multi-modal communication
Exercise	Enables small, independent physical movements	Promotes physical activity through competence and engagement	Supports physical movement through interactive art

By combining the benefits of digital art therapy and gamified activities, Table 3 demonstrates how gamified AI art therapy may satisfy the complex demands of people with dementia. This hybrid method promotes social connection through meaningful connections and creative cooperation, while maintaining safety by offering safe, controlled activities that do not need physical wandering. The hybrid technique combines balanced, interesting tasks that lower stress and establish insights with memory stimulation for cognitive impairment like short-term memory and reasoning. Additionally, by establishing a balance between competence-driven activities and emotional resilience, it improves learning and reasoning while providing a comprehensive framework for cognitive engagement.

A multi-modal communication framework that uses gamified challenges and nonverbal cues to increase interaction is used to mitigate difficulties with language. While gamified AI art therapy encourages independent movements and involvement through interactive, competence-driven activities, the incorporation of physical exercise is particularly notable. All things considered, this hybrid strategy combines the incentive elements of gamification with the therapeutic advantages of digital art to create a more inclusive and successful dementia care intervention.

5. Conclusion

The need for digital art therapy is growing as dementia continues to affect millions of people and traditional therapies struggle with accessibility and efficacy. In this study, the potential benefits of art therapy—augmented by AI and gamified activities—for improving the mental and emotional health of dementia patients were examined. While AI-powered solutions can offer real-time feedback and personalized therapies, gamified activities strengthen therapy's reach and adaptability to each patient's demands. Through non-pharmaceutical interventions, these technology advancements have the potential to revolutionize dementia care by encouraging emotional expression, cognitive stimulation, and increased independence, making art therapy a more accessible and favorable treatment option. Further studies should concentrate on investigating the use of these technologies to guarantee ethical and effectiveness use in dementia treatment.

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